Chapter 3. Common Distributions

Hypergeometric, binomial, multinomial, geometric, negative binomial, Poisson, exponential, gamma, normal

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Conventions

Many distributions below are introduced through a **representation**: $X \overset{d}{=} Z_1 + \cdots + Z_n$ means X has the same distribution as a sum of the indicated variables. Representations make means, variances and mgfs almost free, by Theorem 2.9 and the product rule for mgfs of independent sums. Throughout, Exp(β) and Gamma(α, β) use the *scale* β (mean β and $\alpha\beta$).

3.1 The hypergeometric distribution

Draw n items without replacement from N items of which D are "marked". The number X of marked items drawn has

$$f(x) = \binom{D}{x}\binom{N-D}{n-x}\Big/\binom{N}{n}, \qquad X \sim H(n; N, D).$$

Proposition 3.1

With $p = D/N$: $EX = np$ and $\operatorname{Var} X = \dfrac{N-n}{N-1}\, np(1-p)$.

Proof. Write $X = \sum_{i=1}^{n} I_i$, where I_i indicates that the ith draw is marked. By symmetry $P(I_i = 1) = p$, and for $i \neq j$, $P(I_i = I_j = 1) = \frac{D(D-1)}{N(N-1)}$. So $EX = np$ and

$$\operatorname{Var} X = np(1-p) + n(n-1)\Big(\frac{D(D-1)}{N(N-1)} - p^2\Big) = np(1-p)\Big(1 - \frac{n-1}{N-1}\Big).$$

■

The factor $\frac{N-n}{N-1}$ is the finite-population correction (cf. sampling without replacement). As $N \to \infty$ with $D/N \to p$ and n fixed, $f(x) \to \binom{n}{x}p^x(1-p)^{n-x}$: sampling without replacement from a huge population is like sampling with replacement.

3.2 Binomial and multinomial distributions

Definition (Binomial)

$X \sim B(n,p)$ if $f(x) = \binom{n}{x}p^x(1-p)^{n-x}$, $x = 0, \ldots, n$. Equivalently $X \overset{d}{=} Z_1 + \cdots + Z_n$ with $Z_i \overset{\text{i.i.d.}}{\sim}$ Bernoulli(p).

Proposition 3.2

$M(t) = (1 - p + pe^t)^n$, $EX = np$, $\operatorname{Var} X = np(1-p)$. If $X_i \sim B(n_i, p)$ are independent then $\sum X_i \sim B(\sum n_i, p)$.

Proof. Each Bernoulli has mgf $1 - p + pe^t$, mean p and variance $p(1-p)$; multiply mgfs and add means and variances. For the sum, the mgfs multiply to $(1 - p + pe^t)^{\sum n_i}$, and the mgf determines the distribution. ■

Definition (Multinomial)

Each of n independent trials falls in category j with probability p_j ($\sum_j p_j = 1$). The counts $\mathbf{X} = (X_1, \dots, X_k)$ have

$$f(\mathbf{x}) = \frac{n!}{x_1! \cdots x_k!} p_1^{x_1} \cdots p_k^{x_k}, \qquad \sum x_j = n, \qquad \mathbf{X} \sim \text{Multi}(n, \mathbf{p}).$$

Equivalently $\mathbf{X} \overset{d}{=} \mathbf{Z}_1 + \cdots + \mathbf{Z}_n$ with i.i.d. $\mathbf{Z}_i \sim \text{Multi}(1, \mathbf{p})$ (one-hot vectors).

Proposition 3.3

$X_j \sim B(n, p_j)$; $E\mathbf{X} = n\mathbf{p}$; $\text{Var}\,\mathbf{X} = n(\text{diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}')$, i.e. $\text{Cov}(X_i, X_j) = -np_ip_j$ for $i \neq j$. Merging categories gives again a multinomial.

Proof. For one trial, $\mathbf{Z}$ is one-hot with $E\mathbf{Z} = \mathbf{p}$ and $E(Z_iZ_j) = 0$ for $i \neq j$, so $\text{Var}\,\mathbf{Z} = \text{diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}'$. Sum over n i.i.d. trials. A merged count is a sum of indicators of a union of categories. ■

The negative covariances reflect the constraint $\sum X_j = n$.

3.3 Geometric and negative binomial distributions

Definition (Geometric and negative binomial)

In i.i.d. Bernoulli(p) trials, the trial number X of the first success is Geo(p): $f(x) = (1-p)^{x-1}p$, $x = 1, 2, \dots$. The trial number of the rth success is Negbin(r, p): $f(x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}$, $x = r, r+1, \dots$. Equivalently $\text{Negbin}(r, p) \overset{d}{=} Z_1 + \cdots + Z_r$ with $Z_i \overset{\text{i.i.d.}}{\sim} \text{Geo}(p)$.

Proposition 3.4

For $X \sim \text{Geo}(p)$: $M(t) = \frac{pe^t}{1-(1-p)e^t}$ for $(1-p)e^t < 1$, $EX = \frac{1}{p}$, $\text{Var}\,X = \frac{1-p}{p^2}$. Hence Negbin($r, p$) has mean $\frac{r}{p}$ and variance $\frac{r(1-p)}{p^2}$. Moreover the geometric distribution is **memoryless**: $P(X > m + k \mid X > m) = P(X > k)$.

Proof. The mgf is a geometric series. From $C(t) = \log M(t)$, $C'(0) = 1 + \frac{1-p}{p} = \frac{1}{p}$, and differentiating once more gives $\frac{1-p}{p^2}$. For memorylessness, $P(X > k) = (1-p)^k$, so the ratio is $(1-p)^{m+k}/(1-p)^m$. The waiting times between successive successes are i.i.d. geometric, which gives the representation. ■

3.4 The Poisson distribution

Definition (Poisson)

$X \sim \text{Poisson}(\lambda)$ if $f(x) = e^{-\lambda}\lambda^x/x!$, $x = 0, 1, \dots$.

Proposition 3.5 (Poisson approximation)

If $np_n \to \lambda > 0$, then $\binom{n}{x} p_n^x (1-p_n)^{n-x} \to e^{-\lambda}\lambda^x/x!$ for every x.

Proof. $\binom{n}{x}p_n^x = \frac{n(n-1)\cdots(n-x+1)}{n^x}\cdot\frac{(np_n)^x}{x!} \to \frac{\lambda^x}{x!}$, and $(1-p_n)^{n-x} = (1-\frac{np_n}{n})^n(1-p_n)^{-x} \to e^{-\lambda}$, using $(1+a_n/n)^n \to e^a$ when $a_n \to a$. ■

So for many trials with a small success probability, the number of successes is approximately Poisson. E.g. $B(500, 0.004)$ gives $P(X=0) = 0.996^{500} = 0.135$, against $e^{-2} = 0.135$.

Proposition 3.6

$M(t) = \exp\{\lambda(e^t-1)\}$; $EX = \operatorname{Var} X = \lambda$; independent Poisson($\lambda_i$) variables sum to Poisson($\sum\lambda_i$).

Proof. $\sum_x e^{tx}e^{-\lambda}\lambda^x/x! = e^{-\lambda}e^{\lambda e^t}$. The cgf $\lambda(e^t-1)$ has all derivatives at 0 equal to λ. Mgfs of independent summands multiply. ■

The Poisson process. Let N_t count events in $[0,t]$. Assume (a) stationarity: $N_{s+t} - N_s \stackrel{d}{=} N_t$, and $N_0 = 0$; (b) independent increments over disjoint intervals; (c) $P(N_h = 1) = \lambda h + o(h)$; (d) $P(N_h \geq 2) = o(h)$ as $h \downarrow 0$.

Theorem 3.7

Under (a)–(d), $N_t \sim$ Poisson(λt).

Proof. Let $g(x,t) = P(N_t = x)$. Decompose by the increment on $(t, t+h]$ and use (a), (b):

$$g(x,t+h) = g(x,t)P(N_h=0) + g(x-1,t)P(N_h=1) + o(h) = g(x,t)(1-\lambda h) + g(x-1,t)\lambda h + o(h).$$

Hence $\partial_t g(x,t) = -\lambda g(x,t) + \lambda g(x-1,t)$, i.e. $\partial_t\{e^{\lambda t}g(x,t)\} = \lambda e^{\lambda t}g(x-1,t)$, with $g(0,0) = 1$ and $g(x,0) = 0$ for $x \geq 1$. Solving successively for $x = 0, 1, 2, \ldots$ gives $g(0,t) = e^{-\lambda t}$, then $g(x,t) = e^{-\lambda t}(\lambda t)^x/x!$ by induction. ■

Heuristically: split $[0,t]$ into n pieces; each has an event with probability about $\lambda t/n$, independently, so $N_t \approx B(n, \lambda t/n) \to$ Poisson(λt).

3.5 Exponential and gamma distributions

In a Poisson process with rate λ, the waiting time W_1 to the first event satisfies $P(W_1 > t) = P(N_t = 0) = e^{-\lambda t}$, so it has density $\lambda e^{-\lambda t}$: the **exponential distribution** Exp(β) with mean $\beta = 1/\lambda$. Like the geometric, it is memoryless: $P(W > s+t \mid W > s) = P(W > t)$.

The waiting time W_r to the rth event satisfies $P(W_r > t) = P(N_t \leq r-1)$; differentiating gives the density $\frac{\lambda^r}{(r-1)!}t^{r-1}e^{-\lambda t}$. Allowing a real shape $\alpha > 0$:

Definition (Gamma)

$X \sim$ Gamma(α, β) if $f(x) = \dfrac{1}{\Gamma(\alpha)\beta^\alpha}x^{\alpha-1}e^{-x/\beta}$, $x > 0$, where $\Gamma(\alpha) = \int_0^\infty x^{\alpha-1}e^{-x}\,dx$; α is the shape and β the scale. Exp(β) = Gamma($1, \beta$).

Recall $\Gamma(\alpha) = (\alpha-1)\Gamma(\alpha-1)$, $\Gamma(n) = (n-1)!$ and $\Gamma(\frac{1}{2}) = \sqrt{\pi}$.

Proposition 3.8

For $X \sim$ Gamma(α, β): $EX = \alpha\beta$, $\operatorname{Var} X = \alpha\beta^2$, $M(t) = (1-\beta t)^{-\alpha}$ for $t < 1/\beta$. Independent Gamma(α_i, β) variables (common scale) sum to Gamma($\sum\alpha_i, \beta$). In particular Gamma(r, β) $\stackrel{d}{=} Z_1 + \cdots + Z_r$ with $Z_i \stackrel{\text{i.i.d.}}{\sim}$ Exp(β), which identifies W_r above.

Proof. Substitute $y = x/\beta$: $EX^k = \beta^k\Gamma(\alpha+k)/\Gamma(\alpha)$, giving $\alpha\beta$ and $\alpha(\alpha+1)\beta^2$. Substitute $y = (1/\beta - t)x$ in $\int e^{tx} f(x)\,dx$ to get $(1-\beta t)^{-\alpha}$. Multiply the mgfs for the sum. ■

3.6 The normal distribution

Definition (Normal)

$Z \sim N(0,1)$ has density $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$ and cdf Φ. $X \sim N(\mu, \sigma^2)$ if $X \overset{d}{=} \mu + \sigma Z$, i.e. $f(x) = \frac{1}{\sigma}\phi(\frac{x-\mu}{\sigma})$.

That $\int \phi = 1$ follows from $(\int e^{-z^2/2} dz)^2 = \iint e^{-(x^2+y^2)/2}\,dx\,dy = 2\pi$ in polar coordinates.

Proposition 3.9

(a) $EX = \mu$, $\operatorname{Var} X = \sigma^2$. (b) $M(t) = \exp(\mu t + \frac{1}{2}\sigma^2 t^2)$. (c) $aX + b \sim N(a\mu + b, a^2\sigma^2)$, and independent normals sum to a normal with the means and variances added.

Proof. (b) $Ee^{tZ} = \int \frac{1}{\sqrt{2\pi}} e^{-(z-t)^2/2} e^{t^2/2}\,dz = e^{t^2/2}$ (complete the square); then $Ee^{t(\mu+\sigma Z)} = e^{\mu t} e^{\sigma^2 t^2/2}$. (a) The cgf $\mu t + \frac{1}{2}\sigma^2 t^2$ has first and second derivatives μ and σ^2 at 0. (c) Compare mgfs. ■

Quantiles. The upper α point z_α satisfies $P(Z > z_\alpha) = \alpha$ (e.g. $z_{0.05} = 1.645$, $z_{0.025} = 1.96$), and the upper α point of $N(\mu, \sigma^2)$ is $\mu + \sigma z_\alpha$. Probabilities are computed by standardizing: $P(X \le x) = \Phi((x-\mu)/\sigma)$.

The normal arises as the limit of standardized binomials (de Moivre–Laplace), $P(a \le \frac{X-np}{\sqrt{np(1-p)}} \le b) \to \Phi(b) - \Phi(a)$, a special case of the central limit theorem (Chapter 5).

Summary of representations

distribution	pmf/pdf	representation
$B(n,p)$	$\binom{n}{x} p^x (1-p)^{n-x}$	sum of n i.i.d. Bernoulli(p)
Multi($n, \mathbf{p}$)	$\frac{n!}{\prod x_j!} \prod p_j^{x_j}$	sum of n i.i.d. Multi($1, \mathbf{p}$)
Negbin(r,p)	$\binom{x-1}{r-1} p^r (1-p)^{x-r}$	sum of r i.i.d. Geo(p)
Poisson(λ)	$e^{-\lambda}\lambda^x/x!$	limit of $B(n, \lambda/n)$
Gamma(r, β)	$\frac{x^{r-1}e^{-x/\beta}}{\Gamma(r)\beta^r}$	sum of r i.i.d. Exp(β)
$N(\mu, \sigma^2)$	$\frac{1}{\sigma}\phi(\frac{x-\mu}{\sigma})$	$\mu + \sigma Z$, $Z \sim N(0,1)$